
Effect of Compression Apparel on Variables associated with Performance, Posture, Comfort and Fatigue

Human Performance Laboratory
The University of Calgary
Calgary, Canada

A summary document highlighting the effects of compression apparel on variables associated with performance, posture, comfort and fatigue.

Dourthe, Benjamin
Nigg, Sandro
Lam, Chris
Fletcher, Jared R.
Martinez, Aaron
Omu, Onotobor (Tobi)
Nigg, Benno M.

Biomechanigg Sport & Health
Research Inc.
&
Human Performance Laboratory
University of Calgary
2500 University Drive N.W.
Calgary, Alberta
Canada T2N 1N4

Phone: (403) 220-3436
Fax: (403) 284-3553
E-mail: sandro.nigg@ucalgary.ca

Introduction

Per4ma has developed compression shorts and compression shirts aimed at improving the performance, posture, comfort, and reduce the fatigue of individuals who wear their apparel. In order to determine if the apparel has the desired and intended effects, Biomechanigg Sport & Health Research Inc. was asked to conduct a study with the objective to:

Compare the Per4ma shorts and shirts with a leading competitor and a control condition with respect to its effect on performance, posture, comfort and fatigue.

Methods

Subjects

Data were collected on a total of 20 participants, with 10 subjects having completed the testing for the shirt and 10 subjects having completed the testing of the shorts.

Apparel Conditions

The shirts and short brands that were evaluated in this study included Per4ma, Under Armour, and adidas (Figure 1). Under Armour served as compression control whereas adidas served as a non-compressive control.



Figure 1 Tested shirts and shorts. Left to right: Per4ma, Under Armour, adidas.

Highlights of Testing Protocol

Subjects visited the Human Performance Lab at the University of Calgary on three separate days with a minimum of 48 hours between testing sessions to avoid any effects of fatigue. Upon arrival to the lab, subjects were randomized to testing either the shirt or the shorts. Subjects were asked to carry out a variety of tasks on an isokinetic dynamometer and perform a 30-second Wingate test on either a friction-loaded arm ergometer (arms) or on a magnetically-braked cycle ergometer (legs; Figure 2). Skin temperature, which was used as a surrogate measure of muscle temperature, was measured during a 5-minute warm-up period prior to leg or arm ergometry. Static posture was assessed by an experienced and qualified physical therapist. Upon completion of each test, subjects were asked to fill out a questionnaire regarding their qualitative experiences with the product.



Figure 2 Testing devices used during data collection. Top Left clockwise: isokinetic dynamometer (leg), isokinetic dynamometer (arm), arm ergometer, cycle ergometer.

Variables of Interest

A variety of variables were collected to determine the effects of the apparel with respect to themes of interest (e.g. performance, fatigue, etc.). The variables included:

- Passive torque during shoulder/hip hyper-extension
- Active torque-angular velocity during shoulder/hip flexion
- Arm/leg ergometry (peak power output, fatigue index)
- Shoulder/hip skin temperature (surrogate for muscle temperature)
- Garment compression
- Posture score (shirt condition only)
- Subjective assessment (comfort, support, etc.)

Note: Due to the small sample size, no statistical testing was done on these data sets.

Results

Passive torque during hyper-extension

Figure 3 illustrates the mean peak passive torques recorded with each garment during shoulder (left) and hip (right) hyper-extension for all 20 subjects (ten shirt, ten shorts). The average peak passive torque was 4.4% (shirt) and 6.9% (shorts) higher with the Per4ma garments compared to the non-compressive control.

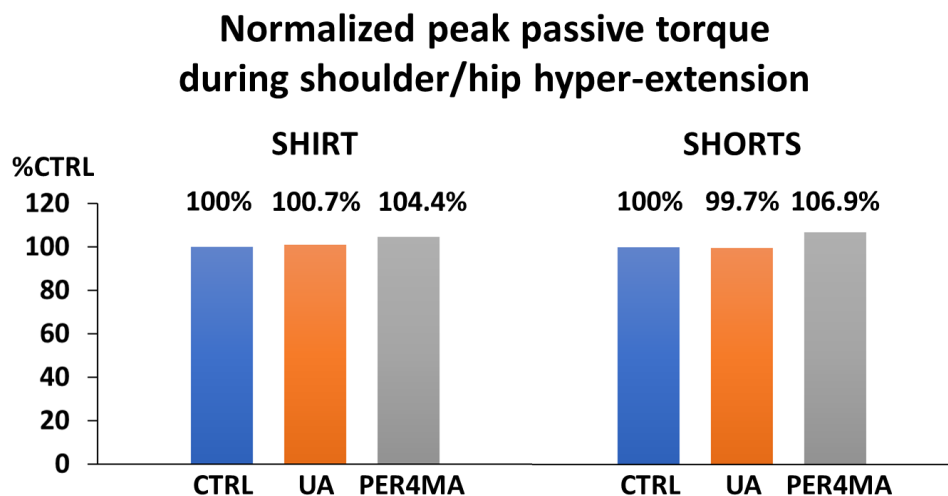


Figure 3 Mean peak passive torque measured during shoulder (left) and hip (right) hyper-extensions for the control (CTRL, blue), Under Armour (UA, orange), and Per4ma (PER4MA, grey) conditions (N = 10 for each garment).

Active torque-Angular Velocity

Figure 4 illustrates the mean peak active torque measured during shoulder (left) and hip (right) flexion for all 20 subjects (10 shirt, 10 shorts) for seven angular velocities (0, 30, 60, 90, 120, 300 and 500 deg/sec). In the shirt condition, the angular velocities of 0, 30, 60, and 500 deg/sec showed higher mean peak active torques in the Per4ma garment compared with the two control garments. In the shorts condition, the Per4ma garment enabled higher mean peak active torques than the two control garments for all angular velocities except 500 deg/sec. The non-compressive control and the Under Armour garments produced very similar peak active torques.

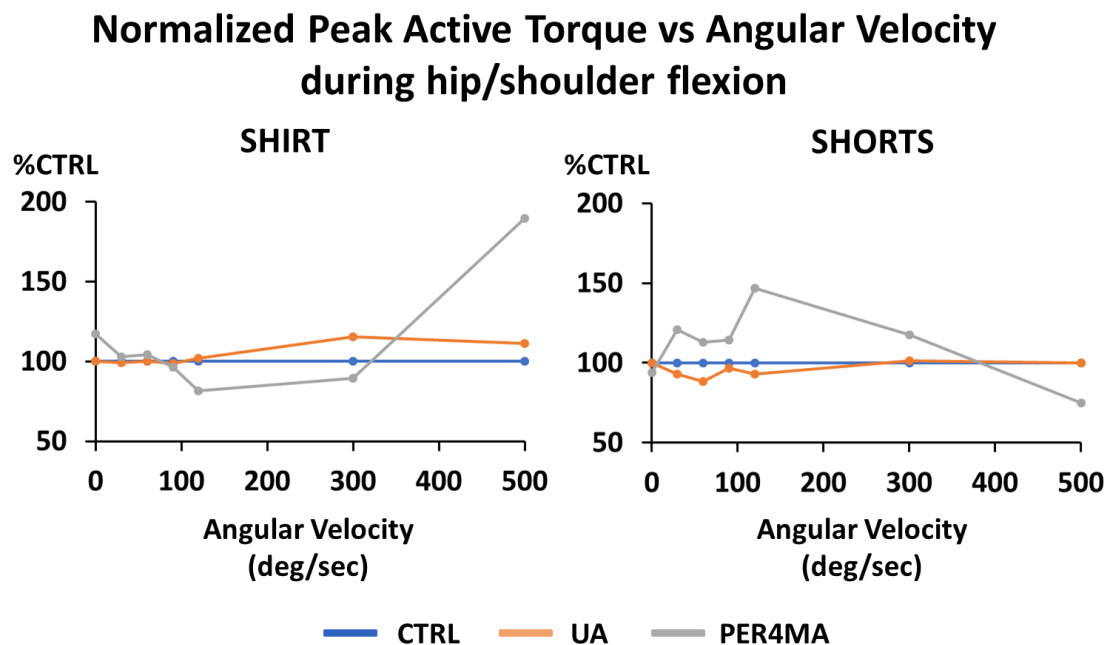


Figure 4 Mean peak active torque measured during shoulder (left) and hip (right) flexion for seven angular velocities (0, 30, 60, 90, 120, 300 and 500 deg/sec) for the control (CTRL, blue), Under Armour (UA, orange), and Per4ma (PER4MA, grey) conditions (N = 10 for each garment).

Ergometry

Figure 5 and Figure 6 illustrate the average peak power and fatigue index calculated during 30-sec arm (left) and leg (right) Wingate trials for 16 subjects (6 shirt, 10 shorts). In the shirt condition, the mean peak power was higher in the Per4ma (4.5%) and Under Armour (9.7%) garments compared to the non-compressive control. No differences in peak power were observed in the shorts conditions.

The mean fatigue index for the shirt comparison increased by 86.2% and 107.4% in the Under Armour and Per4ma shirts, respectively, compared to the compressive control garment. Conversely, when comparing the control shorts with the Under Armour and Per4ma, the mean fatigue index decreased by 8.5% and 7.0%, respectively.

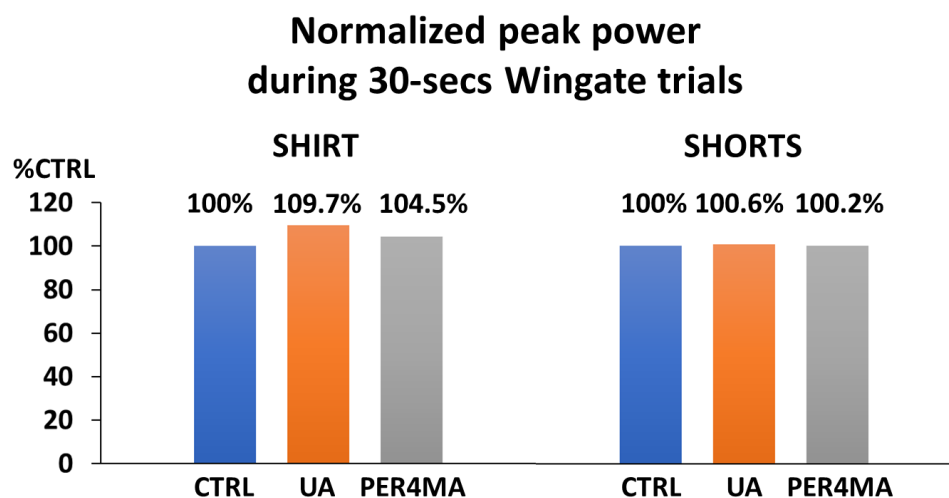


Figure 5 Mean peak power measured during arm (left) and leg (right) 30-secs Wingate trials for the control (CTRL, blue), Under Armour (UA, orange), and Per4ma (PER4MA, grey) conditions (N = 10 for each garment).

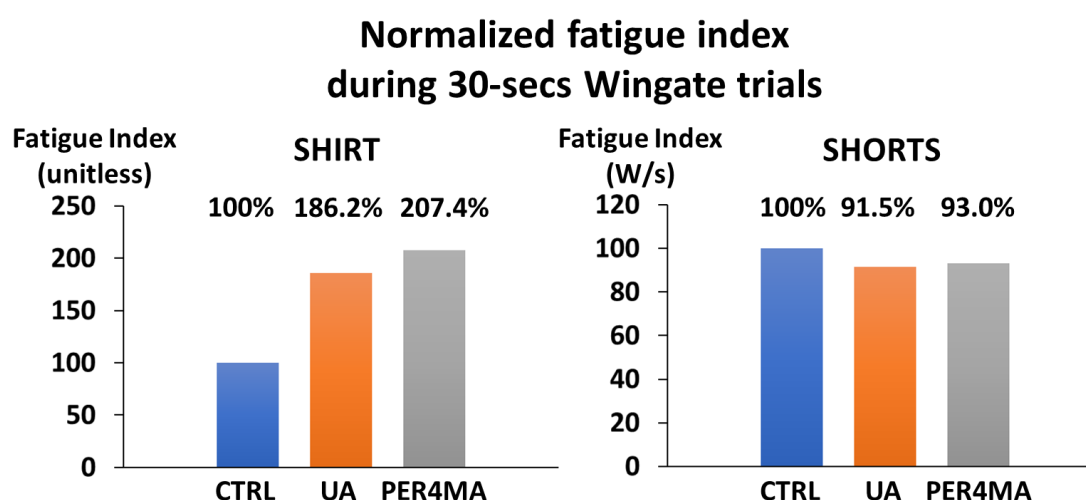


Figure 6 Mean fatigue index measured after arm (left) and leg (right) 30-secs Wingate trials for the control (CTRL, blue), Under Armour (UA, orange), and Per4ma (PER4MA, grey) conditions (N = 10 for each garment).

Skin temperature

Figure 7 shows the evolution of the average skin temperature for the shoulder (left) and hip (right) measured on 20 subjects (10 shirt, 10 shorts) while performing a 5-min warm-up on either the arm or leg ergometer, as well as the skin temperature measured following a 30-sec Wingate test. Results show a slow increase in skin temperature throughout warm-up and post-Wingate trial for all garments and for the shirt and shorts conditions. However, no major difference was found between garments in either condition.

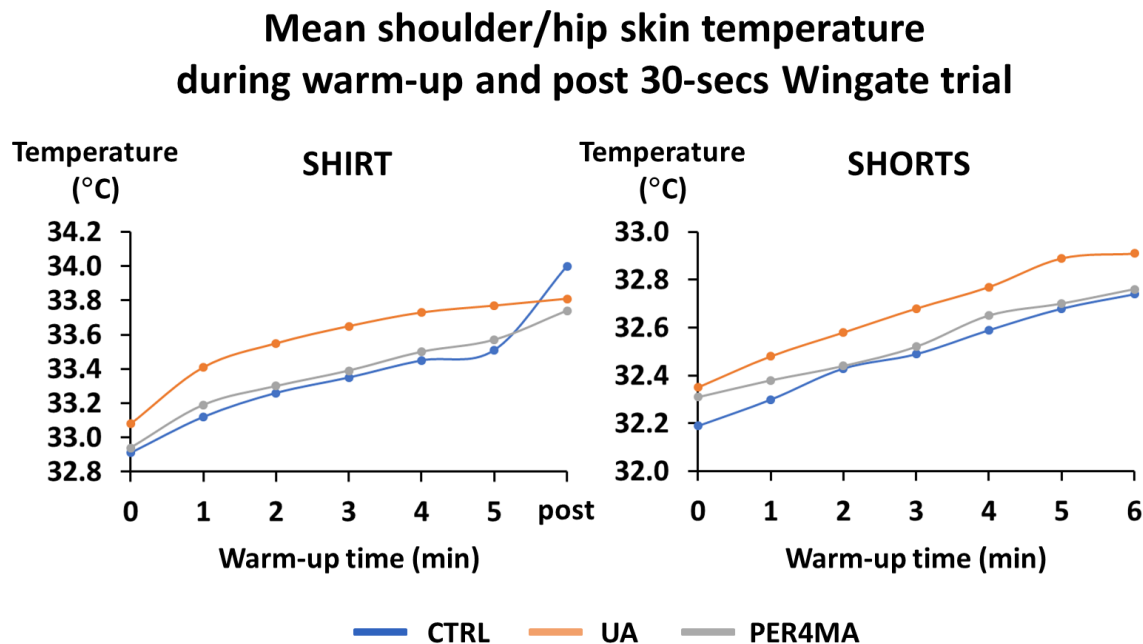


Figure 7 Mean shoulder (left) and hip (right) skin temperature evolution during a 5-min warm-up and post 30-secs Wingate for the control (CTRL, blue), Under Armour (UA, orange), and Per4ma (PER4MA, grey) conditions (N = 10 for each garment).

Garment Compression

Figure 8 illustrates the average increase in garment compression between the Under Armour and Per4ma garments. The compression was higher in the Per4ma garment compared to the Under Armour garment at the biceps, shoulder, thigh and hamstring areas. A smaller increase in compression was measured for the chest and gluteus areas in the Per4ma compared to the Under Armour garments.

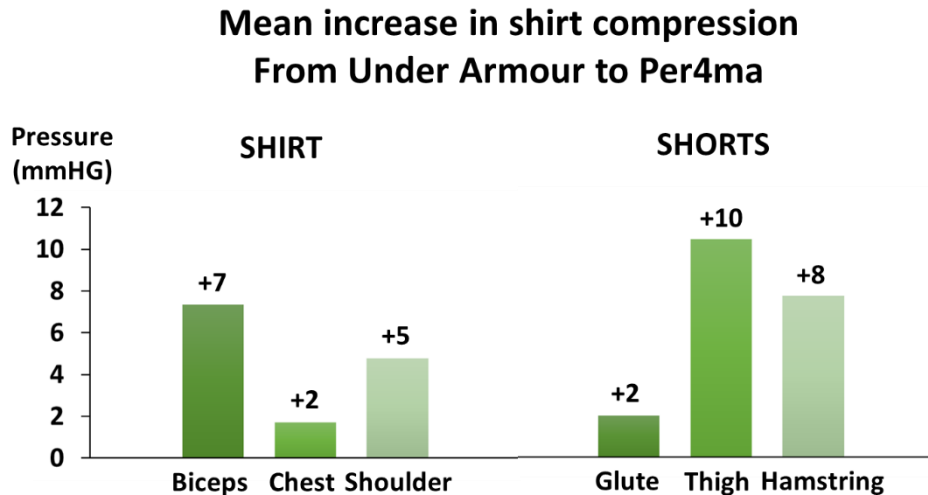


Figure 8 Mean shirt (left) and shorts (right) compression increase from Under Armour to Per4ma.

Posture Score

For the posture assessment, the head, shoulders, spine, hips, ankles, neck, upper back, trunk, abdomen and lower back were assessed by the same qualified physical therapist. The physical therapist rated the head, spine, hips, ankle and abdomen postures as 10 out of 10 across all conditions and subjects, and therefore these results are not included below.

For the remaining six areas, the mean and total (sum) scores across all subjects and conditions are presented in Table 1. The Per4ma shirt had the highest total score and the control the lowest score, however these differences are likely not clinically meaningful with respect to overall posture.

Table 1 Mean and total posture scores for different body segments in the control (blue), Under Armour (orange) and Per4ma (grey) conditons (N = 7).

Body Segment	CTRL	UA	PER4MA
Shoulders	8.0	8.0	8.0
Neck	9.0	9.5	10.0
Upper back	8.5	9.5	10.0
Trunk	9.0	10.0	10.0
Lower back	9.5	9.5	10.0
Total score (out of 50)	44.0	46.5	48.0

Subjective Assessment

The results from the subjective questionnaires are shown in Tables 2 to 4. Seventy percent of the participants found the Per4ma shirt to be comfortable, and 86% found the Per4ma shorts to be comfortable. Most participants found the Per4ma garments to be the most supportive. Approximately 30% of the subjects indicated that they would be willing to wear the Per4ma shirt to recover from a shoulder injury or to increase their athletic performance. For the shorts condition, 29% and 57% of the population were willing to wear the Per4ma garment to recover from a hip injury and increase their athletic performances, respectively. Knowing that this study only looked at the impact of the Per4ma shorts during biking, future work should look at additional tasks (e.g. running, jumping, skating). Such investigation may highlight more task-specific trends reflecting people's willingness to wear the Per4ma garment after an injury or to enhance performances.

Table 2 Percentage of subjects who found the garments comfortable, supportive and who would be willing to wear them to recover from an injury or to improve their athletic performance.

	SHIRT			SHORTS		
	CTRL	UA	PER4MA	CTRL	UA	PER4MA
Was the garment comfortable	100%	100%	70%	100%	100%	86%
Was the garment supportive	0%	50%	90%	0%	57%	57%
Would you wear this garment to recover from a shoulder/hip injury	0%	20%	30%	14%	29%	29%
Would you wear this garment to improve your athletic performance	20%	30%	30%	14%	71%	57%

The Per4ma garments (shirt and shorts) were considered tighter than the non-compressive control and Under Armour garments. Most participants found the Per4ma garments tighter around the biceps, chest, shoulders and thighs (Table 3). Most participants found that the Per4ma garments made it harder to perform passive shoulder and hip hyper-extensions compared to the control and Under Armour garments. Similar perceptions were observed for the active shoulder and hip flexions, where the movements were rated as more difficult in the Per4ma compared to the control.

Table 3 Compression assessment in both shirt and shorts conditions. Scale ranged from zero (too tight) to 15 (too loose).

	SHIRT			SHORTS		
	CTRL	UA	PER4MA	CTRL	UA	PER4MA
Biceps/Glute compression	10.8	6.9	3.0	12.1	7.4	6.0
Chest/Thigh compression	10.5	6.6	5.5	12.9	7.7	2.7
Shoulder/Hip compression	11.5	6.7	4.6	12.4	6.8	6.3

No major difference was found between garments regarding perception of fatigue after performing the 30-sec Wingate trial for either the shirt or shorts conditions. Most participants felt warmer while wearing the Per4ma garments compared to the control and Under Armour garments.

Table 4 Assessments of (1) ability to perform passive and active movements, (2) perception of fatigue after performing the 30-sec Wingate trial, and (3) temperature perception while wearing the garments.

	SHIRT			SHORTS		
	CTRL	UA	PER4MA	CTRL	UA	PER4MA
Passive movement assessment (0: very easy – 15: very hard)	3.6	6.1	8.0	0.6	3.2	6.5
Active movement assessment (0: very easy – 15: very hard)	4.3	6.3	5.6	2.0	4.5	5.9
Fatigue level after Wingate (0: exhausted – 15: energized)	3.4	3.9	3.6	2.3	2.5	3.5
Temperature perception (0: too hot – 15: too cold)	7.9	6.4	5.2	7.8	6.6	5.0

Discussion

In the present study, three compression shorts and shirts were evaluated with respect to variables associated with performance, fatigue, skin temperature, garment compression, posture, and subjective perceptions. From this data set, some small differences between garments can already be observed and the below points highlight the benefit to be expected when wearing the Per4ma garment.

The Per4ma garments provided more resistance during passive shoulder and hip hyper-extension. It is speculated that this added resistance may limit the range of motion for these two joints, and therefore limit the risks of hyper-extension related injuries.

The Per4ma shirt provided more resistance during slow and fast shoulder flexion movements, which may contribute to an improved control of shoulder mobility, thereby limiting the risks of shoulder injuries.

The Per4ma shorts provided more resistance during slow and moderate hip flexion movements, which may contribute to an improved control of hip mobility, thereby limiting the risks of hip injuries.

The Per4ma shirt resulted in a positive impact on overall body posture, specifically at the neck, upper back and lower back areas.

The Per4ma shirt was rated as more supportive and more willing to be worn to recover from a shoulder injury compared to the competitor products.

The Per4ma garments were rated as tighter around the biceps, shoulders, hips and thighs, which may explain the lower overall comfort rating.